



Translating Life Sciences

Kenya Institute of Primate Research

POLICY DOCUMENT

Information and Communications Technology (ICT) Policy

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FOREWORD



The ICT Policy of KIPRE is a strategic blueprint designed to harness the power of technology in advancing the institution's mandate of biomedical research. Recognizing the pivotal role of information and communication technologies in modern research, this policy outlines a comprehensive framework to guide the acquisition, utilization, and management of ICT resources. It emphasizes the

importance of secure, reliable, and efficient technological infrastructure to support cutting-edge research, data analysis, and collaboration within the institution and with external partners.

This policy is underpinned by relevant Kenyan legislation, including the Kenya Information and Communications Act, the Data Protection Act, and the Computer Misuse and Cybercrimes Act. It also aligns with international best practices and standards in information security and data protection. By adhering to these legal and ethical frameworks, the institution ensures that its ICT operations are conducted responsibly, protecting sensitive research data and upholding the privacy and security of all stakeholders.

This policy demonstrates KIPRE's commitment to innovation and staying at the forefront of scientific discovery. It promotes the integration of technology into all aspects of research activities, from data collection and analysis to knowledge dissemination and collaboration. By prioritizing a technologically advanced environment, the institution aims to accelerate scientific progress, enhance research outcomes, and ultimately contribute to the betterment of human health in Kenya and beyond.

Signed:

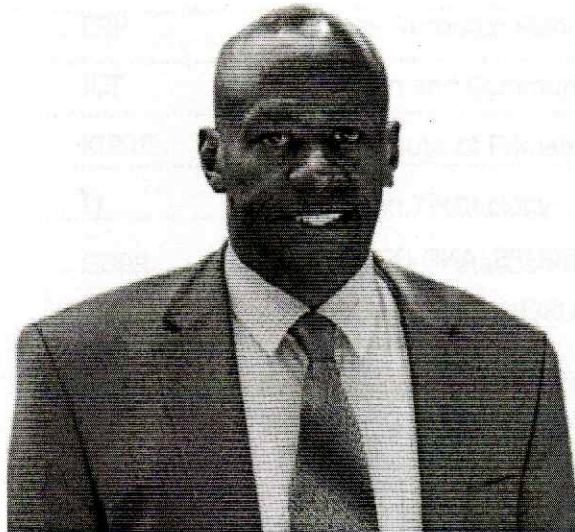
Josephat Mulimba

Professor Josephat Mulimba, PhD., FRSM
Chairman, Board of Directors

Date: 10 day of May 2024



PREFACE



The Kenya Institute of Primate Research (KIPRE) has developed this policy document as a necessary intervention to address use of Information Communication Technology (ICT) resources, including the ICT network equipment, access to outside networks through the local network, software, and digital media.

This policy addresses and strives on the commitment to facilitate the digitization and digitalization of KIPRE's research activities.

Using innovative technologies, we aim to equip our employees with the skills and knowledge necessary to influence existing research technology effectively. By providing adequate and reliable systems, we attempt to increase the reach of our publications, e-books, journals, and articles, fostering an environment of knowledge sharing thus enabling continuous collaboration and access to a wealth of knowledge resources.

We cannot overlook the significance of information security in a world growing more interconnected by the day. Thus, we are committed to ensuring the integrity of our data and enhancing the continuity of our operations in the face of unforeseen disasters through regular backups and vigorous security protocols.

Let us seize the chances that lie ahead and work together to create a future in which ICT acts as a catalyst for development and advancement.

Signed:



Dr. Peter Gichuhi Mwethera, PhD, MBS
Ag. Director General

Date: 10 day of MAY 2024

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ABBREVIATIONS

LAN	Local Area Network
ERP	Enterprise Resource Planning
ICT	Information and Communications Technology
KIPRE	Kenya Institute of Primate Research
IT	Information Technology
GDPR	General Data Protection Regulation
NTP	Network Time Protocol
DCS	Director Corporate Services
CSD	Corporate Services Directorate

DEFINITION OF TERMS

User is any user, employee, student, contractor, agent, temporary worker, vendor, and any other person in a position to know or obtain information about computers or devices on the Local area Network (LAN).

Firewall is a hardware or software device which protects the ports of computers on the LAN. For the purposes of this document.

Remote Access shall mean access to the Local Area Network from any location outside the firewall by any method, including but not limited to Virtual Private Network (VPN), dial-in modem, frame-relay, cable-modem and any other method of accessing the LAN from outside the firewall.

Information and Communications Technology (ICT) Asset means all software; hardware and data used in the management of the related KIPRE information resources. (Note: This may include non-ICT resources (e.g. Printed records).

Information System means an electronic system that manages information and data related to the ICT Asset.

Information Technology (IT) - Is "a fancy name for data processing", according to Newton, IT means all equipment, processes, procedures and systems used to provide and support information systems (both computerized and manual) within an organization and those reaching out to customers and suppliers.

Internet is the world-wide collection of private and public router-based networks that are interconnected via gateways and exchange points, which all utilize the TCP/IP protocol.

IP address is a set of protocols developed to allow cooperating computers to share resources across a network.

Local Area Network (LAN) is a computer network that spans a relatively small area such as a single building or group of buildings.

Network Time Protocol (NTP) is an internet protocol that allows computers to synchronize their clocks over packet-switched networks.

Backup The saving of files onto offline mass storage media for the purpose of preventing loss of data in the event of equipment failure or destruction.

Archive The saving of old or unused files onto magnetic tape or other offline mass storage media for the purpose of releasing on-line storage room.

Restore The process of bringing offline storage data back from the offline media and putting it on an online storage system such as a file server.

ARTICLE 1: INTRODUCTION

The Kenya Institute of Primate Research (KIPRE), whose mandate is preclinical biomedical research, is a semi-autonomous government agency established under the Legal Notice No. 273 of 2017 in the Science, Technology and Innovation Act No. 28 of 2013. KIPRE, operates under the Ministry of Health, State Department for Public Health and Professional Standards through the Presidential Executive order No. 1 of January, 2023.

1.1 Vision

To be a global center of excellence in innovation, basic, preclinical, biomedical research non-human primate conservation

1.2 Mission

To improve human health and biodiversity through advances in basic and preclinical biomedical research and non-human primate conservation

1.3 Core Values

The core values include:- Research integrity, Professionalism, Good Corporate Governance, Equity, Customer Focus

1.4 Guiding Principles

The guiding principles include:- Efficiency, Teamwork, Partnership, Innovativeness, Dynamism

1.5 Functions

The functions (as stated in legal notice No. 273 of 2017) of the Institute shall be to-

- a) Conduct basic, pre-clinical and translational research using non-human primates and other experimental animals in liaison with the relevant government agencies;
- b) Establish an incubation centre linking research and industry in the health products value chain;
- c) Contribute to early warning systems for emerging and reemerging diseases;
- d) Establish and operate a national breeding colony of nonhuman primates and other experimental laboratory animals for research purposes;
- e) Provide national in-vitro fertilization services for endangered wildlife of Kenya in liaison with relevant Government agencies;
- f) Undertake studies in non-human primate conservation biology and Primatology;
- g) Create and maintain a national non-human primate database on census, research affiliations, and geo-spatial distribution of non-human and related zoonotic diseases;
- h) Serve as a national research centre for venom and anti-venom including snake bite disease;
- i) Undertake basic and pre-clinical research on reproductive health, infectious and non-

- communicable diseases using non-human primates;
- j) Develop and characterize both non-human primate and other experimental animal models for priority diseases in Kenya;
 - k) Undertake research in Human African Trypanosomiasis disease;
 - l) Undertake capacity building in animal welfare, ethical use and care of non-human primates and other experimental animals in laboratory settings; and
 - m) Perform any other function which is incidental to the performance of any of the foregoing functions.

ARTICLE 2: POLICY STATEMENT, RATIONALE, SCOPE, AND OBJECTIVES

2.1 Policy statement

- 2.1.1 KIPRE is committed to promote effective communication and working practices within the organisation using Information and communication technologies (ICT) systems.
- 2.1.2 KIPRE is committed to protect information resources against events that may jeopardise information security by contaminating, damaging, or destroying information resources.
- 2.1.3 KIPRE is committed to keep information technology (IT) policies current and relevant.

2.2 Rationale

The ICT Policy shall ensure that the ICT infrastructure and capacity are utilized effectively and are in alignment with the KIPRE's strategic objectives, National ICT Policy, National e-Government Strategy and the e-Government Standards and Guidelines.

2.3 Scope

- 2.3.1 The Policy applies to any person granted authorization to access any computer or device, software, systems and email on the Institute's LAN (an "Authorized User") or via remote access. This includes but not limited to contractors, temporary workers, vendors, sub-contractors, and partners.
- 2.3.2 This policy covers all IT facilities, hardware, software, and services provided by the Institute. These are:

a) Facilities:-

- i. Computer labs
- ii. Server room(s)
- iii. ICT maintenance room
- iv. All ICT facilities installed at KIPRE

b) Services

- i. Provision of guidance and expertise training on ICT
- ii. ICT support in software, hardware and any other computing infrastructure
- iii. Technical support to staff and students

c) Hardware

- i. PCs, Laptops and servers
- ii. Printers and Scanners
- iii. Network routers and switches
- iv. Power backup equipment (e.g. Uninterruptable Power Backup - UPS)
- v. L.C.D Projectors
- vi. Network Devices
- vii. Cameras (Digital and Camcorders)
- viii. PDAs, Smartphones and other Mobile Computing Devices
- ix. Diskettes/CDs/DVDs
- x. Flash-disks/external hard-disks
- xi. PABXs, Telephone heads, fax and photocopiers
- xii. All other ICT related hardware

d) Systems

- i. Enterprise Resource Planning (ERP) System
- ii. Institutional Repository
- iii. Research and Grants Management System
- iv. Information management system
- v. Data capture tools and databases
- vi. Any other system processes that may be considered appropriate

e) Software

- i. Network operating software
- ii. PC operating software
- iii. Research software
- iv. Utility software
- v. Custom made software

2.4 Objectives of the policy

- i. To facilitate digitization and digitalization of KIPRE research activities
- ii. To promote dissemination of research outputs (publications, e-books, journals, articles) through provision of adequate and reliable systems
- iii. To promote efficient utilization of ERP system for improved transaction processing and more accurate data entry and reporting.
- iv. To ensure information security and enhance continuity of operations in the event of a disaster through regular backups.
- v. To utilize cost-effective training methods to ensure the full use of existing information technology capabilities by employees.

ARTICLE 3: ADMINISTRATIVE ACCESS TO LAN USER'S INFORMATION

The authorizations and restrictions listed below are intended to help LAN administrators fulfil their duties while safeguarding LAN users' privacy.

3.1 Authorizations

LAN administrators are particularly permitted to obtain, keep, and utilise the following categories of data kept in computers connected to KIPRE LAN for the purposes of system management and maintenance:

- 3.1.1 LAN administrators are permitted unlimited access to a network user's computer when that user has given explicit consent for such access, as during installation, upgrade, troubleshooting, and repair operations.
- 3.1.2 LAN administrators have read-only access to information about the current configuration of any user's networked computer, subject to the limitation that no user-input information is to be collected by the administrator. In practice, this authorization would allow the administrator to use software to identify the user's CPU, keyboard, and monitor types, amounts of free RAM and disk space, operating system and version, available ports, and similar useful configuration data such as that contained in generic configuration files.
- 3.1.3 LAN administrators have unlimited read and write access to one specified directory and its subdirectories on the networked computer and one specified individual-user-drive directory and its subdirectories on each attached server, subject to the limitations described later in this paragraph.

3.1.4 LAN administrators are permitted to make back-up copies of all files stored on servers.

3.1.5 LAN administrators are permitted to use software that identifies and directs specific file types on the network to specific network bandwidth areas to protect the maximum amount of network bandwidth to support research activities that require internet connectivity.

3.2 Restrictions

KIPRE shall be responsible for ensuring that electronically stored information is treated as confidential, with the exceptions mentioned above.

ARTICLE 4: USER PRIVILEGE

User privileges indicate the types of authorization issued to different LAN users as follows.

4.1 Local Computer Privileges

There are three main categories of users on a computer or network. These categories include:

- 4.1.1 **Restricted user** - Can operate the computer and save documents but can't alter system settings.
- 4.1.2 **Standard user (power user)** - Can change system settings and install programs that don't affect Windows system files.
- 4.1.3 **Administrators** - Have complete access to read and write any data on the system and add or remove any programs or change system settings.

4.2 Network Privileges

The following categories of network resources shall be available to the users as outlined below.

- 4.2.1 **Email** - Users shall have full access to their own institutional email and shall not be able to transfer ownership to someone else.
- 4.2.2 **A personal network drive on a networked file server** - The user shall not be able to transfer ownership to someone else.
- 4.2.3 **A shared group or organizational division's drive** - Access may be read or write and may vary by organizational requirements.
- 4.2.4 **Access to databases** - Database administrators shall only have full access to the database that they administer.

4.3 Additional Privileges

Groups that may be allowed additional access include:

1. **Domain administrator** - Has full privileges on all computers in the domain including servers and workstations. Privileges include reading and writing data, installing programs, and changing settings.
2. **Server operator** - Has full privileges on servers including reading and writing of data, installing programs, and changing settings.
3. **Backup operator** - Allowed to read data on the domain for the purpose of saving files to backup media. This group cannot write all data on the domain.
4. **Account operator** - Can manage and view information about user accounts on the domain.

ARTICLE 5: USE OF PASSWORD

To safeguard network security, data integrity, and computer systems, staff who have access to organisational computer systems are required to abide by the password policies listed below.

5.1 Password Protection

5.1.1 All Authorized Users must use strong passwords. Unacceptable passwords include but are not limited to;

- i. first or last names, or combinations thereof;
- ii. names of an Authorized User's children or pets;
- iii. words found in a dictionary, combinations of dictionary words with a sound alike digit (four4, etc);
- iv. use of the words or variants on the word password, admin, update, access, login, computer, terminal, workstation, work, home, etc.
- v. Names of people or places as part of your password.
- vi. Part of your login name in your password.

5.3.2 Authorized Users should change their password regularly.

5.3.3 No User shall leave a password written down in proximity to the computer or device which the password accesses.

5.3.4 No User shall provide their login or email password to anyone, including family members.

5.2 Password Requirements

The following password requirements shall be set by the ICT division:

1. Minimum Length - 8 characters recommended
2. Maximum Length - 14 characters
3. Minimum complexity - Passwords shall use three or four of the following four types of characters:
 - i. Lowercase
 - ii. Uppercase
 - iii. Numbers
 - iv. Regular expressions such as !@#\$%^&*(){}[]
4. Passwords are case sensitive and the username or login ID is not case sensitive.
5. Password history - Require a number of unique passwords before an old password may be reused.

6. Maximum password age - 60 days
7. Minimum password age - 2 days
8. Store passwords using reversible encryption with special authorization by the ICT division.
9. Account lockout threshold - 4 failed login attempts
10. Reset account lockout after 20 minutes.
11. Account lockout duration – In some circumstances such as possible break in attempts the account lockout shall be between 30 minutes and 2 hours.
12. Password protected screen savers should be enabled and should protect the computer within 5 minutes of user inactivity.

ARTICLE 6: ANTI-VIRUS

6.1 The organization shall protect its gadgets from malicious software by use of anti-virus product.

The following guidelines shall be applied in line with best practices and applicable laws

- i. The anti-virus product shall be operated in real time on all servers and client computers.
- ii. The product shall be configured for real-time protection.
- iii. The anti-virus library definitions shall be updated at least once per day.
- iv. Anti-virus scans shall be done a minimum of once per week on all user-controlled workstations and servers.

6.2 The email server shall have additional protection against malware since email with malware must be prevented from entering the network.

6.3 Email Malware Scanner shall scan emails as soon as they arrive at the server and before they depart. Once a week, the scanner may check every email that has been saved for malware or viruses. If malware or a virus is discovered, the email shall be deleted without notification.

6.4 Any other malware including but not limited to adware and spyware shall be prevented and removed.

6.5 Computers operating outside the organizational network have a local firewall software program operational at all times when these computers are connected to the internet.

ARTICLE 7: EMAIL & INTERNET CONNECTION

7.1 Email Use

7.1.1 Staff shall receive Institutional email addresses that shall be used for all official correspondence.

7.1.2 Email users shall refrain from sending broadcast messages. Email shall be sent to intended audience.

7.1.3 KIPRE mail service shall not be used to disseminate unofficial requests or information.

7.1.4 KIPRE shall respect users' privacy and promote the use of email according to KIPRE data management protection and sharing policy and Data Protection Act (2019)

7.1.5 KIPRE staff shall not subscribe to un-official websites using Institutional email address

7.1.6 KIPRE staff shall use Institutional email for sharing conferencing links

7.2 Physical Internet Connection

All internet connections to other private networks shall be authorized and approved by the Director Corporate Services. These connections include but not limited to:

7.2.1 Modem connection from a computer or communication device which may allow a

connection to the network.

7.2.2 Any multipurpose printing and FAX machines which have both a phone and network connection.

7.2.3 Wireless access points or devices with wireless capability.

7.3 Internet Control and Logging System

All documents, messages and attachments composed, sent, received or stored on electronic mail or Internet/LAN storage systems remain the property of KIPRE.

A system shall be required to operate on the network with the following capabilities:

7.3.1 The ability to prevent users from visiting inappropriate websites.

7.3.2 The ability to log user internet activity including:

- i. Time of the internet activity.
- ii. Duration of the activity.
- iii. The website visited.
- iv. Data and type of data downloaded
- v. Whether the system will cache web pages to increase the internet connection speed. This requires a proxy server.

7.3.3 The system (shall / shall not) require a login ID or it will use the current network login to identify users.

7.4 Right to Monitor.

The institute's electronic mail and LAN systems, along with any attachments created, received, or sent, websites visited, and/or files downloaded, are all subject to review, audit, intercept, access, and/or disclosure rights.

7.5 Internet Site/Identification Originator.

Employees should be aware that Internet sites accessed from KIPRE's computer network may identify the institute as the originator of each visit. If employees participate in "chat sessions" or post messages on the Internet, they may be regarded as representing the institute. Thus, all communications must be professional, appropriate to KIPRE, and not adversely reflect upon its reputation.

ARTICLE 8: USE OF SOFTWARE

8.1. Software Purchasing

All software shall be purchased through ICT Support with the written agreement of the System Administrator. This includes any external purchases such as Web Hosting / sites, online tools, etc.

8.2 Workstations and Laptops

Software shall not be installed, loaded or run on any ICT equipment that has not been approved for installation by the System Administrator. Staff who wish to have software installed on their office computers shall submit a request in writing/email to the Deputy Director ICT.

8.3 Licensing and Copyright

All users of computer facilities shall be expected to understand and comply with the copyright laws as they apply to computer software and its documentation, and to refrain from creating or using illegal copies of software on the Institute's computing equipment.

KIPRE's Software use provides additionally that:

1. Genuinely licensed software shall be installed on computers.
2. All software and data files stored on Institutional computer equipment shall be consistent with the Institute's Acceptable Use.
3. Installed software shall fall into one of the following categories:
 - i. Is in the public domain.
 - ii. Is covered by a licensing agreement with the software author, authors, vendor or developer, whichever is applicable.
 - iii. Has been donated to the college by the author, authors, vendor or developer and a written record of the contribution is on file in the office of the System Administrator.
 - iv. Has been purchased by the Institute and a record of that purchase is on file with the System Administrator.
 - v. Has been purchased by the user, approved for installation by ICT Support, and a record of a that purchase exists and can be produced by the user upon demand.
4. Archival or backup copies of application or data files shall be made only as specified by applicable license agreements.

8.4 Hacking

KIPRE has a “Zero Tolerance” when it comes to the making, supplying or possession of any software that can be used for unauthorized access into computer networks and other ICT systems.

Any breach of this will lead to disciplinary action being taken against those staff or students involved.

8.5 Institute Supported Software

A list of Software supported by the Institute is available on request from the System Administrator.

Only legitimately acquired software may be used and only in accordance with all applicable license conditions.

A software register (to include software name, serial number of product, date of purchase and the location of software) should be established and maintained to enable verification of software compliance.

ARTICLE 9: APPLICATION IMPLEMENTATION

9.1 This shall be used to evaluate how new applications may affect security. To maintain the stability of the network, the impact of newly developed apps that offer users or internal groups new capabilities must be evaluated.

9.2 Users shall work together with application developers and computer security experts to assess data requirements for any new applications. Users shall specify their requirements for the applications and application developers will work with the user to identify and categorize data according to the Application Development Security Assessment Process.

9.3 Once the data and application requirements are established, computer security personnel can then evaluate risk and determine methods, processes, equipment, and procedures to mitigate known risks. The computer security personnel, users, and application developers will work together to provide required and reasonable access capability to systems and data.

9.4 The data assessment, risk evaluation, and system requirements should be done early in the project life cycle since without this information, the overall cost of the project cannot be accurately assessed.

ARTICLE 10: SYSTEMS MANAGEMENT

KIPRE systems shall be managed as follows;

- i. KIPRE's Information Systems shall be managed by suitably trained and qualified staff to oversee their day-to-day running and to preserve data security, confidentiality and integrity.
- ii. All systems management staff must have up to date Information Security and General Data Protection Regulation (GDPR) training and retake this training at suitable regular intervals.
- iii. System Administrators responsible for Information Systems are to maintain the appropriate access controls for their systems and keep records of any elevated access they give to users.
- iv. System Administrators or individuals responsible for Information Systems are responsible for correct and secure operation of computers in accordance with related KIPRE policies.
- v. Access to all Information Systems, excluding publicly accessible data sources, shall use a secure authentication process. Consideration should also be given as to whether it is appropriate and feasible to further limit or control access to business-critical systems.
- vi. Access to information systems is to be logged and monitored where appropriate.
- vii. Access to all Information Systems, excluding publicly accessible data sources, shall use a secure authentication process. Consideration should also be given as to whether it is appropriate and feasible to further limit or control access to business-critical systems. Access to information systems is to be logged and monitored where appropriate.
- viii. System Administrators and individuals responsible for the Information Systems must ensure that appropriate backup and system recovery procedures are in place, dependent upon the assessed level of criticality of the information concerned. Backup of KIPRE's information systems and critical assets and the ability to recover then is an important priority. Recovery from backup should be regularly tested.
- ix. System Administrators should, where appropriate and feasible, include resilience and redundancy features. These need to be regularly tested to confirm they function as designed and suitable records kept of the test result.
- x. System Administrators are responsible for ensuring that the KIPRE's systems and critical data is frequently backed up and procedures for recovery meet the needs of the business.
- xi. Only authorized staff will be permitted to perform systems administration functions. Use of commands to perform these functions should be logged and monitored where it is considered appropriate and feasible to do so.
- xii. Formal change control procedures, with audit trails, shall be used for all changes to business-critical systems. All such changes must be risk assessed and authorized by the Director General or relevant line director/ICT Deputy Director before being moved to the live environment.

- xiii. Systems connected to the network should have appropriate configuration and controls, such as firewalls, to defend the system from inappropriate access. Only required services should be offered by the system.
- xiv. Remote access to services should be appropriately secure, controlled, authenticated (inc MFA) and accounted for by design and not rely on additional measures such as VPN.
- xv. System clocks must all be synchronized to KIPRE's Network Time Protocol (NTP) service. In the case of computers in the Active Directory this will happen automatically.

ARTICLE 11: ASSET CONTROL

All employees and personnel that have access to organizational computer systems shall adhere to the IT asset control in order to protect the security of the network, protect data integrity, and protect and control computer systems and organizational assets. The asset control shall not only enable organizational assets to be tracked concerning their location and who is using them but it will also protect any data being stored on those assets.

11.1 Assets Tracked

This section defines what IT assets shall be tracked and to what extent they should be tracked.

- i. Desktop workstations
- ii. Laptop mobile computers
- iii. Printers, Copiers, FAX machines, multifunction machines
- iv. Handheld devices
- v. Scanners
- vi. Servers
- vii. Firewalls
- viii. Routers
- ix. Switches

11.2 Asset Tracking Requirements

- i. All assets must have a Serial Number. Either an internal tracking number will be assigned when the asset is acquired or the use of Manufacturer ID numbers must be specified.
- ii. An asset tracking database shall be created to track assets. It will include all information on the Asset Transfer Checklist table and the date of the asset

change.

- iii. When an asset is acquired, an ID will be assigned for the asset and its information shall be entered in the asset tracking database.

11.3 Asset Transfers

This applies to any asset transfers including the following:

- i. Asset purchase
- ii. Asset relocation
- iii. Change of asset trustee including when an employee leaves or is replaced.
- iv. Asset disposal

11.4 Asset Disposal

Asset disposal is a special case since the asset must have any sensitive data removed prior to disposal. Below is listed the action for the device based on data sensitivity according to the data assessment process.

- i. None (Unclassified) - No requirement to erase data but in the interest of prudence normally erase the data using any means such as reformatting or degaussing.
- ii. Low (Sensitive) - Erase the data using any means such as reformatting or degaussing.
- iii. Medium (Confidential) - The data must be erased using an approved technology to make sure it is not readable using special hi technology techniques.
- iv. High (Secret) - The data must be erased using an approved technology to make sure it is not readable using special hi technology techniques. Approved technologies are to specified in a Media Data Removal Procedure document by asset type including:
 - a. Floppy disk
 - b. Memory stick
 - c. CD ROM disk
 - d. Storage tape
 - e. Hard drive.
 - f. RAM memory
 - g. ROM memory or ROM memory devices.

ARTICLE 12: BACKUP

This article defines the backup for computers within the organization which are expected to have their data backed up.

12.1 Timing

Full backups shall be performed at 11.pm daily

12.2 Media Storage

There shall be separate sets of storage media

12.3 Responsibility

The ICT Deputy director shall delegate a member of the ICT division to perform regular backups. The delegated person shall develop a procedure for testing backups and test the ability to restore data from backups.

12.4 Testing

The ability to restore data from backups shall be tested at least once per month.

12.5 Data Backed Up

Data to be backed up include the following information:

1. User data stored on the hard drive.
2. System state data
3. The registry

Systems to be backed up include but are not limited to:

1. File server
2. Mail server
3. Production database server
4. Domain controllers
5. Test database server

12.6 Archives

Archives are made at the end of every financial year in July. User account data associated with the file and mail servers are archived one month after they have left the organization.

12.7 Restoration

Users that need files restored shall submit a request to the help desk. Include information about

the file creation date, the name of the file, the last time it was changed, and the date and time it was deleted or destroyed.

12.8 Media Storage Locations

Offline storage media used for daily backup shall be stored in a fireproof safe.

ARTICLE 13: NETWORK & SERVER DOCUMENTATION

Network and server documentation defines the level of network documentation required such as documentation of which switch ports connect to what rooms and computers, who shall have access to read network documentation and who shall be notified when changes are made to the network.

13.1 Network Documentation

The network structure and configuration shall be documented and provide the following information:

1. IP addresses of all devices on the network with static IP addresses.
2. Server documentation on all servers as outlined in the "Server Documentation" document.
3. Network drawings showing:
 - i. The locations and IP addresses of all hubs, switches, routers, and firewalls on the network.
 - ii. The various security zones on the network and devices that control access between them.
 - iii. The locations of every network drop and the associated switch and port on the switch supplying that connection.
 - iv. The interrelationship between all network devices showing lines running between the network devices.
 - v. All subnets on the network and their relationships including the range of IP addresses on all subnets and netmask information.
 - vi. All wide area network (WAN) or metropolitan area network (MAN) information including network devices connecting them and IP addresses of connecting devices.
4. Configuration information on all network devices including:

- i. Switches
 - ii. Routers
 - iii. Firewalls
- 5. Configuration shall include but not be limited to:
 - i. IP Address
 - ii. Netmask
 - iii. Default gateway
 - iv. DNS server IP addresses for primary and secondary DNS servers.
 - v. Any relevant WINS server information.
- 6. Network connection information including:
 - i. Type of connection to the internet or other WAN/MAN including T1,T3, frame relay.
 - ii. Provider of internet/WAN/MAN connection and contact information for sales and support.
 - iii. Configuration information including netmask, network ID, and gateway.
 - iv. Physical location of where the cabling enters the building and circuit number.
- 7. DHCP server settings showing:
 - i. Range of IP addresses assigned by all DHCP servers on all subnets.
 - ii. Subnet mask, default gateway, DNS server settings, WINS server settings assigned by all DHCP servers on all subnets.
 - iii. Lease duration time.

13.2 Server Documentation

For every server on a secure network, there are a list of items that shall be documented and reviewed on a regular basis to keep the institute network secure. This list of information about every server shall be created as servers are added to the network and updated regularly.

- 1. Server name
- 2. Server location
- 3. The function or purpose of the server.
- 4. Hardware components of the system including the make and model of each part of the system.
- 5. List of software running on the server including operating system, programs, and

services running on the server.

6. Configuration information about how the server is configured including:
 - i. Event logging settings
 - ii. A comprehensive list of services that are running.
 - iii. Configuration of any security lockdown tool or setting
 - iv. Account settings
 - v. Configuration and settings of software running on the server.
7. Types of data stored on the server.
8. The owners of the data stored on the server.
9. The sensitivity of data stored on the server.
10. Data on the server that should be backed up along with its location.
11. Users or groups with access to data stored on the server.
12. Administrators on the server with a list of rights of each administrator.
13. The authentication process and protocols used for authentication for users of data on the server.
14. The authentication process and protocols used for authentication for administrators on the server.
15. Data encryption requirements.
16. Authentication encryption requirements.
17. List of users accessing data from remote locations and type of media they access data through such as internet or private network.
18. List of administrators administrating the server from remote locations and type of media they access the server through such as internet or private network.
19. Intrusion detection and prevention method used on the server.
20. Latest patch to operating system and each service running.
21. Groups or individuals with physical access to the area the server is in and the type of access, such as key or card access.
22. Emergency recovery disk and date of last update.
23. Disaster recovery plan and location of backup data
24. Mail Server Documentation
 - i. Account size limit where the person receives warnings about mailbox size

- ii. Account size limit where the person cannot send mail anymore.
- iii. Account size limit where the person cannot receive mail anymore.

13.3 Change Notification

The Deputy director ICT shall notify staff when the following network changes are made;

1. Reboot of a network device including switches, routers, and firewalls.
2. Changes of rules or configuration of a network device including switches, routers, and firewalls.
3. Upgrades to any software on any network device.
4. Additions of any software on any network device.
5. Changes to any servers which perform significant network functions whether configuration or upgrade changes are made. These servers include:
 - i. DHCP
 - ii. DNS
 - iii. Domain controllers
 - iv. WINS

Notification shall be through email to designated groups of people.

13.4 Documentation Review

The Deputy director shall ensure that network documentation is kept current by performing a monthly review of documentation or designating a staff member to perform a review.

13.5 Storage Locations

Network documentation shall be kept either in written form or electronic form in a minimum of two places. It should be kept in two facilities at least two miles apart so that if one facility is destroyed, information from the other facility may be used to help construct the ICT infrastructure. Information in both facilities should be updated monthly at the time of the documentation review.

ARTICLE 14: IT EQUIPMENT PURCHASE AND FAILURE PREVENTION

14.1 Equipment Requirements

All critical services are required to utilize redundant technologies including:

1. Dual power supplies on all servers providing critical services.
2. RAID disk arrays to prevent one disk failure from interrupting services
3. Uninterruptable power supplies that can provide power for a minimum of 1 hour to servers operating critical services in the event of a power outage.

14.2 Additional Requirements

For services that are critical for income or operations that cannot be interrupted the following technologies are also recommended:

1. A backup generator to ensure that long term power outages cannot interrupt service.
2. More than one server for the same service where the servers use clustering or load balancing technology.

14.3 Critical Services

Critical services which are required for normal operation of the organization include:

1. File sharing service on a file sharing server.
2. Web services to the internet
3. Email services
4. Database services for internal users and critical external applications.
5. Critical external application servers.
6. Domain controller servers
7. Firewall to connect these services to the internet.

Any servers or equipment that supports these services should adhere to this including connection equipment from the internet to these services.

ARTICLE 15: SERVER MONITORING

Server monitoring applies to all production servers and infrastructure support servers including but not limited to the following types of servers:

1. File servers
2. Database servers
3. Mail servers
4. Web servers
5. Application servers
6. Domain controllers
7. FTP servers
8. DNS servers

15.1 Daily Checking

All servers shall be checked manually on a daily basis. the following items shall be checked and recorded:

1. The amount of free space on each drive shall be recorded in a server log.
2. The system log shall be checked and any major errors shall be checked and recorded in the server log.
3. Services shall be checked to determine whether any services have failed.
4. The status of backup of files or system information for the server shall be checked daily.

15.2 External Checks

Essential servers shall be checked using either a separate computer from the ones being monitored or a server monitoring service. The external monitoring service shall have the ability to notify multiple IP personnel when a service is found to have failed. Servers to be monitored externally include:

1. The mail server
2. The web server
3. External DNS servers
4. Externally used application servers.
5. Database or file servers supporting externally used application servers or web servers.

ARTICLE 16: USER TRAINING

Given the current state of computer security and the fact that many attacks are directly against the user's web browser or through e-mail, user training is becoming an ever more important part of computer security. User training does not need to be extremely technical but should ensure that the user retains basic computer knowledge along with some knowledge about specific computer attacks that they may experience.

16.1 Training Categories

Training categories shall include but not be limited to the following areas:

S/No	Training Category	Key areas
1.	Basics	i. What files are ii. How to set view for details and show extensions for known file types iii. Why not seeing file extensions is a security hazard to you iv. How to determine file storage size v. Mail attachments vi. Where to store files; how to use your network drive and what your network drive is and what it means to you vii. How to copy files viii. Ways to increase efficiency on the computer such as keyboard shortcuts
2.	Email use	i. How to set up email for remote users or with your ISP with POP3/IMAP ii. How to set up out of office reply iii. How to set mail filtering rules iv. How to use, import, and export personal folders v. What an undeliverable response to an email message means
3.	Use of web browser	i. Safe browser? ii. Avoid adware and spyware - ignore ads that may compromise your computer or get you to install an

		illicit program iii. How to change browser settings for better security iv. Products to prevent malware.
4.	Ways to get malware	i. Through email ii. Through browser iii. By connecting iv. By installing unapproved programs
5.	Email Viruses	i. How they spread ii. Spoofing sender iii. Dangerous attachments
6.	Email Spam	i. Protect your email address ii. Filtering spam
7.	Hoaxes	i. Phishing ii. Fraud methods
8.	Passwords	i. Why protect my password? ii. Why do I need to change my password every 30 days iii. How to change your password iv. How to choose strong passwords that you can remember v. If I log in on a website can someone see my password?
9.	Other	i. Reasons for firewall -- worms and others ii. Why worry about malware? iii. What is a vulnerability? iv. Why not run all services? v. Social engineering

16.2 Requirements

All organizational staff shall make measurable and continuous progress in the training areas. Each employee manager shall be responsible for ensuring that employees under their supervision make progress in the required training areas.

ARTICLE 17: COMPUTER MAINTENANCE

17.1 Need for Maintenance

The following are reasons why computers need maintenance:

1. Microsoft critical security updates need to be applied at least once a month
2. Firewall, virus and spyware protection needs regular review and management
3. Daily use of servers and workstations by office staff can create random network issues
4. Management of backup status, data selected and data testing is critical to data recovery
5. Proactive server and workstation standardization minimizes problems
6. Management of hard drive resources prevents storage issues & server crashes
7. Analysis of server event logs can identify issues before they create network problems
8. Regular optimization of server(s) and workstations to increase speed and efficiency
9. Proactive maintenance provides real peace of mind that someone is regularly caring for your network
10. Predictable monthly budget and support minimizes financial and technical surprises

Because of these and other factors, computer require regular computer service maintenance to keep the big problems away and maintain optimal performance.

The forms in the appendices are designed to assist the system administrator maintain the computer systems in their day-to-day operations.

17.2 Computer lifespan

As hardware ages, it might not meet the performance requirements of users or new OSes and software, potentially leading to a decrease in productivity. There is need for a systematic strategy to manage a computer's entire lifespan within an organization. This includes purchase planning, set up, maintenance and, ultimately, disposal of these machines as per the procurement policy on disposal of assets.

IT equipment shall be disposed if;

- i. A computer/laptop/mobile has been in use for a period of 5 years
- ii. The cost of repairing damaged IT hardware is comparable to that of buying a new one.
- iii. The IT equipment is no longer working and cannot be salvaged
- iv. The IT equipment is no longer supporting KIPRE's current needs

17.3 Misuse of Computing Resources

The following actions constitute misuse of the Institute's computer resources and are strictly prohibited for all Users:

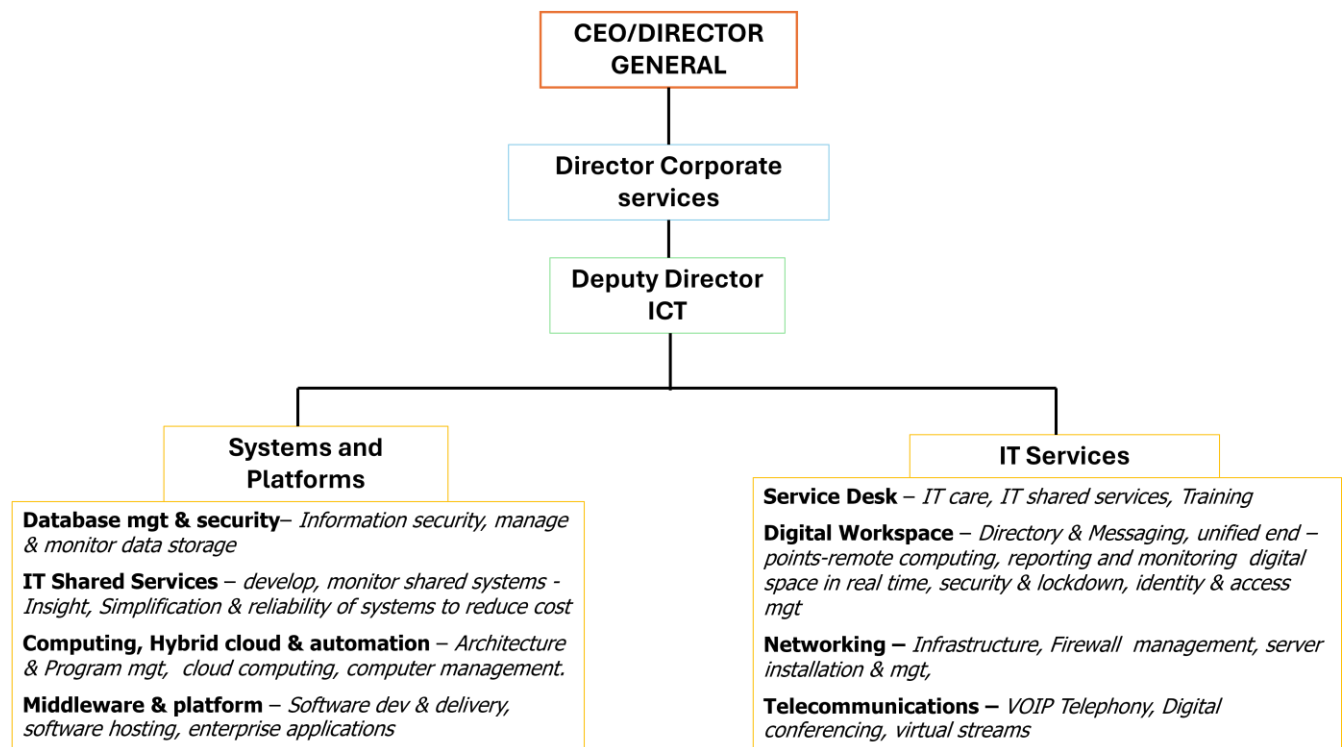
1. Criminal and illegal acts. Institute computer resources are not to be used in support of or for illegal activities. Any such use will be reported and dealt with by the appropriate Institute authorities and/or law enforcement agencies. Criminal and illegal use may involve, but is not limited to, unauthorized access, intentional corruption or misuse of computer resources, theft, obscenity, and child pornography.
2. Failure to comply with laws, policies, procedures, license agreements, and contracts that pertain to and limit the use of the Institute's computer resources.
3. Abuse of computer resources including, but not limited to, any act which endangers or damages specific computer software, hardware, program, network or the system as a whole, whether located on campus or elsewhere on the global Internet; creating or purposefully allowing a computer malfunction or interruption of operation; injection of a computer virus on to the computer system; sending a message with the intent to disrupt Institute operations or the operations of outside entities; print outs that tie up computer resources for an unreasonable time period to the detriment of other authorized users; computing tasks that consume an unreasonable amount of communications bandwidth either on or off campus to the detriment of other authorized users; and failure to adhere to time limitations which apply at particular computer facilities on campus.
4. Use of Institute computer resources for personal financial gain or a personal commercial purpose.
5. Failure to protect a password or account from unauthorized use.
6. Permitting someone to use another's computer account or using someone else's computer account.
7. Unauthorized use, access, reading, or misuse of any electronic file, program, network, or the system.
8. Unauthorized use, access, duplication, disclosure, alteration, damage, misuse, or destruction of data contained on any electronic file, program, network, or Institute hardware or software.
9. Unauthorized duplication and distribution of commercial software and other copyrighted digital materials. All commercial software and many other digital materials are covered by a copyright of some form. The unauthorized duplication and distribution of software and

other copyrighted materials (including copyrighted music, graphics etc) is a violation of copyright law and this policy. Exceptions to this are specific authorization by the copyright holder or use under the fair use provisions of the copyright law.

10. Attempting to circumvent, assisting someone else or requesting that someone else circumvent any security measure or administrative access control that pertains to Institute computer resources.
11. Use of the Institute computer system in a manner that violates other Institute policies such as racial, ethnic, religious, sexual or other forms of harassment.
12. Use of the Institute's computer system for the transmission of commercial or personal advertisements, solicitations, promotions, or employees' transmission of political material that is prohibited

ARTICLE 18: ICT DIVISION

It is hereby established that the Deputy Director ICT Division shall be accountable to the Director Corporate Services (DCS).



The Role of the Division

- 1) Formulate, implement and review information communication technology policies, strategies, procedures, standards and guidelines;
- 2) Advise the Institute on matters Information Communication Technology;
- 3) Ensure compliance with all statutory and regulatory government targets in respect of ICT;
- 4) Maintain, monitor and update ICT infrastructures in the Institute;
- 5) Spearhead the digitization of Institutes records;
- 6) Review business user requirements to automate the business operations and improving the systems;
- 7) Ensure timely backup of Institutes information and records;
- 8) Develop technological solutions and seek partnerships with relevant stakeholders for integration and implementation;

- 9) Plan, develop, review and implement information communication technology corporate disaster recovery and business continuity strategy;
- 10) Provide specifications and standards in the procurement of all information technology hardware, software and services;
- 11) Provide technical support services to Information Communication Technology users and train them to acquire skills needed to implement computer- based information systems;
- 12) Ensure management, protection and security of data;
- 13) Maintain and update information communication technology hardware and software;

ARTICLE 19: LIABILITY

Any criminal liability accruing in relation to this policy shall be criminally handled. Staff who Damage/lose ICT equipment/facilities, shall be required to pay forthwith. KIPRE management may take any other measures as per existing Labour Laws.

ARTICLE 20: RIGHT TO PRIVACY

KIPRE shall be committed to the right of privacy and private use as per the guidelines outlined in the KIPRE data management policy

ARTICLE 21: COMPLIANCE WITH THE POLICY

Any failure on the part of an employee of KIPRE to act in accordance with the Policy shall be reported to the ICT Deputy director who will escalate the matter to the Director General for action.

ARTICLE 22: REVIEW AND AMMENDMENT

This policy shall be reviewed on a regular basis to accommodate demands arising from advancement in research. Under normal circumstances the policy will be reviewed every three years. However, nothing shall limit KIPRE inherent power to review this policy at any time as shall be deemed to be necessary and equitable.

ARTICLE 23: REFERENCES

1. Copyright Act (CAP. 130)
2. Data Protection Act (No.24 of 2019)
3. Media Act, 2013
4. Kenya Information and Communication Act 2013

APPENDIX 1: Computer Maintenance Request form

KENYA INSTITUTE OF PRIMATE RESEARCH (KIPRE)

KIPRE/CSD/ICT/001

COMPUTER SYSTEMS MAINTENANCE REQUEST (To be used to report any faults with computer systems, internet and network or software related problems)	Requested by: _____ phone no/email: _____	Date: _____
	Location of computer: Office: _____ Other: _____	
AREAS OF REQUEST (Check all that apply)		
☐ Computer not powering on ☐ Mouse faulty/moves too quickly(slowly) ☐ Monitor not displaying ☐ Keyboard not working ☐ No sound from the speaker ☐ Printer problems ☐ Computer too slow	☐ No internet connection ☐ No network/ network cable seems unplugged ☐ Disk/flash disk not being detected ☐ Unable to send/ receive emails/email setup error ☐ UPS power issue/noisy etc	☐ Windows has crashed (blue screen) ☐ Virus infection/No antivirus software ☐ Data/ file lost ☐ Windows out of date ☐ Cannot log to ERP/GIS/Network folder ☐ ERP Error Message ☐ Other

FOR official use only- Details problem

Keyboard ☐	Hard Drive ☐	Email ☐	Other Hardware: _____
mouse ☐	Printer ☐	Network/internet ☐	
Cables/power ☐	Monitor ☐	Office/ other software ☐	Routine maintenance (e.g. Antivirus, Backup, Defragmentation, Restore etc): _____
memory ☐	CD/DVD Drive ☐	ERP ☐	

Type of request ☐ repair ☐ Upgrade ☐ Other (Specify) _____

Action Taken: _____

Signature (IT/AIT) _____ Date: _____

User comment

APPENDIX 2: Computer Maintenance Schedule

	Computer Maintenance Schedule sample											
Task	January	February	March	April	May	June	July	August	September	October	November	December
Download Virus Updates	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily
Incremental Backup	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily
Run Cleanup	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily
File Management	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily
Perform Full Virus Scan	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily
Run Spyboot and Adware	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily	Daily
Organize All Programs Menu	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly
Perform Full Backup	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly
Run Start Inspector	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly	Weekly
Check Windows Updates	X	X	X	X	X	X	X	X	X	X	X	X
Uninstall Unsued Programs	X	X	X	X	X	X	X	X	X	X	X	X
Clean Registry	X	X	X	X	X	X	X	X	X	X	X	X
Clean up Hard Disk	X	X	X	X	X	X	X	X	X	X	X	X
Update Drivers/Programs	X			X			X			X		

Clean Keyboard/Mouse	X			X			X			X		
Clean Monitor	X			X			X			X		
Defrag Hard Disk	X	X	X	X	X	X	X	X	X	X	X	X
Run AIDA32 (Print Report)	X			X			X			X		
Clean Inside CPU	X			X			X			X		